

Anthony Ibarra | Electrical & Computer Engineer

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Professional Profile

Electrical and Computer Engineering graduate looking for the opportunity to become a professional in my respective career. Showcasing a strong focus in Embedded Systems, Control theory, Robotics, including pattern recognition involving AI CNN machine learning. And power electronics making me a flexible Embedded Software Engineer with a strong foundation. I possess various knowledge in different fields providing fast adaptable capability within the work environment of a multidisciplinary team. With the eagerness to learn on and off the job, demonstrating reliability and trustworthiness to depend on one another.

Core SKILLS

- | | | |
|-------------------|--------------|--------------|
| • Soldering | • Creativity | • Motivation |
| • Testing | • Designing | • Management |
| • Troubleshooting | • Planning | |
| • Problem Solving | • Leadership | |

Education & Qualifications

University of Illinois at Chicago UIC, Chicago IL

Jan 2023 - May 2025

Masters of Science in Electrical & Computer Engineering, ECE

Relevant Coursework:

- | | |
|---------------------------------------|-----------------------------------|
| • Audio signal processing | • Electromagnetic Compatibility |
| • Intro to filter Synthesis | • Advanced Computer Communication |
| • Adaptive Digital Filters | • Convex Optimization |
| • Electromechanical energy Conversion | • Mechatronics Embedded Design |
| • Linear Systems Theory & Design | |

- Intro Neural Networks

University of Illinois at Chicago UIC, Chicago IL

Aug 2017 - May 2022

Bachelors of Science in Computer Engineering, ECE

Relevant Coursework:

- Artificial Intelligence I
- Data Structures
- Foundation of Computing
- Principles of Modern Control
- Principles of Auto Control, Robc & Control
- Embedded Systems
- Comp organization
- Advanced comp architecture
- Intro VLSI design
- Pattern Recognition I
- Computer Comm Networks I
- Probability and Random Process for engineers

Projects

Jan 2025 - May 2025

Audio & Acoustic Signal Processing

Research Paper(project)

Implement a solution for motorcycle riders with some degree of hearing impairment. The starting goal is to remove the dampening from the helmet, providing clear sound while wearing the helmet without adding any unneeded noise or unwanted sounds. In turn this can potentially amplify user awareness while riding or in largely populated areas. Report being in the scope and style of a Signal Processing Society conference paper.

- Develop a new method or solution with concepts to be applied to an existing problem. Perform real world testing and validation of project and research implementation
- Test and use skills and algorithms from acoustic & audio signal processing to achieve the set goal

Aug 2024 - Dec 2024

Electromechanical Energy Conversion

Series Brush DC Motor Modeling

From a given Datasheet of a motor replicate the graphs of speed, current, power, and efficiency vs. load torque. Use Ltspice to model the motor Brush DC Motor 14207S007-SP, in an electrical circuit and simulate to obtain the graphs.

- Implement a behavior model of a DC motor using knowledge magnetic circuits and transformers; DC machines; and three-phase AC synchronous and induction machines

- DM – 100A DC Machine 0.3 kW (0.4 hp), 125 V, 2.4 A, and 1800 RPM; motor testing and verification, working with motors of up to 120v using 0 – 125 volt variable and 0 – 150 volt DC voltmeter /ammeters
- AC 120VA 60Hz transformer testing through AC measurements using wattmeters, voltage meters, and ammeters
- LTSpice design and simulation of DC motor mechanical behavior through electrical simulation

Jan 2024 - May 2024

Electromagnetic Compatibility Project

Optimization on Three Coil Long Range, Wireless Power Transfer

Based on the provided research papers attempt to recreate the results from said paper. Using Matlab, and HFSS. Provide a professional typeset report regarding the results with Latex.

- Electromagnetic Compatibility testing using a spectrum analyzer, Oscilloscopes for EMC and EMI testing for PCB and signal integrity using RF SMA coaxial cables, SMA terminators
- Create a three coil configuration for long range wireless power transfer for simulation, confirming data from research papers through simulations
- Mathematically simulated plot from Matlab regarding the Three coils as shown within the research paper
- Simulation and plot from Ansys HFSS 3D EM simulation software, simulating the three coil behavior in different configurations
- Make an IEEE formatted report using Latex, in a professionally typeset manner demonstrating discoveries and results

Jan 2023 - May 2023

Mechatronics Embedded Design Project

Self Driving Car

Lead a team of 4 to design and develop an RC car to follow a line on a track. The car includes a DC motor for the four wheel drive, Servo for steering, and a Line camera for track detection.

- Lead the team and manage all time constraint tasks, development and design tasks, make timely decisions for the team's success in completing the product and hardware development
- Design a motor driver circuit through FET Driver implementation, either of (single fet, half bridge, or full H-bridge), controlled via a PWM input signal from the microcontroller
- Design a Boost Converter for the implementation multiple power systems
- Product testing using Oscilloscope, multimeters, benchtop powersupplies; and Soldering fixes when necessary

- Develop and tune controllers for both the steering and velocity controls (Servo motor, DC motor) respectively
- Design, develop the PCB through Altium Designer and get it manufactured, providing all necessary files such as BOM, gerber files, etc.
- Solder all through hole and SMT components including masking for PCB and Prototype; create any needed jumper wires for connections through crimping and soldering
- Prototype implementation of a perf board circuit, also used as an emergency backup board for PCB failure ensuring continuation of product development
- Implement Sensors and Encoders a filter for the line Camera or velocity measured input through embedded software development and test through debugging microcontroller software

Jan 2023 - May 2023

Intro to Artificial Neural Networks

Wildfire Classification

Design a neural-network-based wildfire/no wildfire classification, using a programming language (Python, Matlab, C++, etc.), and any framework(Pytorch, TensorFlow, Keras, Caffee, Matlab neural network toolbox, etc.). Dataset containing 10k images provided, the training and testing are split 9:1 respectively.

- Design a CNN (convolutional neural network) for pattern recognition and classification of real life cases
- Implement classification training for wildfires as it is similarly done for facial recognition
- Configure and tune the training and testing for the cnn model using python and pytorch with cuda cores and GPU's when able
- Use the created image classification model for wildfire prevention in testing

Jan 2022 - May 2022

Robotics: Algorithm/Control

Mobile Robot Control

Program a Raspberry Pi alpha bot to complete tasks within a respective area. Programming language of choice is Python, with the robot containing many actuators and sensors.

- Embedded software and firmware development for the robot task completion
- Design kinematics for robot control and awareness for motion planning
- Implement video monitoring through onboard actuated camera

- Path awareness through implementation of arrays of infrared sensors; including robot localization through Aruco markers located around the environment
- Accomplish specific tasks within a constrained location

Aug 2021 - May 2022

UIC Senior Design

Automated Watering System

Work in a four-student team to prototype a device that water specified plants by taking moisture levels, outdoor weather conditions, and plant information into account.

- Work in a four-student team to prototype a product implementing a automated watering systems plants by taking moisture levels, outdoor weather conditions, and plant information data into account
- Manage team through verbal and written communication to make sure all assignments and goals are completed on time, and submit weekly reports based on progress of project development
- Embedded programming for a Arduino Nano iot 33 to consider watering decisions per plants moisture levels and water consumption
- Create mobile APP (Kivy framework) to display necessary information regarding the system, plants moisture levels and weather conditions
- Create a UDP client-server connection for communication between powered product and the mobile application; providing manual control over a wireless connection
- Implement wireless communication through Bluetooth Protocol (BLE) for consumer and their device

Jan 2022 - May 2022

Pattern Recognition I

Face recognition, Principal Component Analysis (PCA)

Using Python scikit-learn datasets and matplotlib train and test, for face recognition.

- Using a large dataset from MNIST, adapt the dataset for training and testing scenarios
- Tune and adjust training phase accordingly while taking the training and testing results into account

Volunteer

Aug 2017 - May 2025

Table Tennis Club member

TTC at UIC recreation center

Assist with opening, closing, and setting up equipment for club activities. Help with practice and training for future tournaments or events when possible.

2014 - 2017

Volunteer (volunteered 3 times)

HighSchool Heroes Program

The process of teaching 1st-3rd graders about society through a specified curriculum composed into four sections. Volunteers are placed in a group of 3 to complete said curriculum in a concise time frame for each section.

Additional

Interests:

- | | | |
|---------------------------|---------------|---------------------|
| ● Artificial Intelligence | ● Robotics | ● Embedded & |
| ● Machine Learning | ● Automation | Control systems |
| ● Deep Learning | ● Research | ● Power Efficiency |
| ● Neural Networks | ● Development | ● Cost Efficiency |
| ● Programming | ● Design | ● Hardware Security |

Hardware used:

Equipment:

- | | | |
|----------------------|--------------------|-----------------|
| ● Oscilloscope | ● PowerSupply | ● Keysight |
| ● Multimeter | ● Soldering Tools | N9912A(FieldFox |
| ● Function Generator | ● Breadboard | RF analyzer) |
| ● Waveform Generator | ● Microcontrollers | |

MCU:

- | | |
|-----------------------------|----------------------------|
| ● Tiva C launchpad TM4C123G | ● Raspberry pi (alpha bot) |
| ● Arduino Nano 33 IoT | ● Freescale FRDM-KL25Z |

Programming Languages:

- Python
- C/C++
- Assembly
- Verilog
- VHDL
- PLC(ladder logic)
- HMI (SCADA)
- CSS
- Html
- JavaScript
- Rust

Software used:

Programming:

- VS Code
- GitHub
- Git
- Ubuntu(WSL)
- Docker
- Spyder
- Microsoft Word, PowerPoint, Excel

Embedded:

- Code-Composer studio
- Kinetis design studio
- Arduino IDE
- Mars 4_5
- VNC viewer (raspberry Pi)
- STM32 CubeMX
- DMK Extraction tool
- ControlFlash

Circuit design:

- Altium Designer
- Quartus
- Flux
- LTspice

Simulation & Design or Development:

- MATLAB
- Solidworks
- Mathematica
- Design Spark
- Blender
- HFSS Ansys

PLC-HMI-SCADA:

- RSLogix Micro
- RSLinx Classic
- RSLogix Emulate 500
- Connected Component Workbench CCW
- CODESYS

Database:

- mySQL
- SQL

Test Automation:

- Robot Framework
- Selenium

Hobbies:

- Repairing (disassembling and reassembling electronic parts / devices)
- Table-Tennis, video games, DIY projects